

Failure Modes of Belief-Propagation Decoding on Degenerate Quantum LDPC Codes

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Abstract

This note quantifies *why* belief propagation (BP) decoding fails on degenerate quantum LDPC codes. Across exhaustive studies of small codes and large-sample studies of surface-code patches and bivariate bicycle codes, 98–100% of the BP+OSD failures that an exact minimum-weight decoder could have avoided are degenerate ambiguities — syndromes whose minimal errors span several logical classes — while classical stopping sets contribute zero. We calibrate the fractional solutions of the Gu–Soleimanifar LP relaxation as a detector of these ambiguities. A syndrome with two minimal errors always places a fractional point in the optimal face of the LP, so the detector reaches 100% recall by construction, and its precision decays with code distance from 48% at $d \leq 3$ to 9–21% at $d = 4$ –6: degeneracy *without* ambiguity is the sole source of false positives. Finally, the $[[144,12,12]]$ bivariate bicycle code essentially never converges to a wrong answer, while $[[72,12,6]]$ does so on roughly 40% of its BP failures.

1 Introduction

Belief propagation with ordered statistics decoding (BP+OSD) is the default decoder for quantum LDPC codes, including the bivariate bicycle codes of Bravyi et al. [1]. It is known — from the SymBreak analysis of syndrome symmetry [2] and from the LP-decoding analysis of Gu and Soleimanifar [3] — that *degeneracy* is the root cause of BP’s poor behavior on such codes: a syndrome is degenerate when several minimal-weight error patterns explain it, and decoding fails when those patterns disagree on the logical outcome. What remains under-documented is the quantitative split: how much of the avoidable failure is degenerate ambiguity versus classical stopping sets versus plain non-convergence, and whether the fractional solutions of the decoding LP relaxation predict those failures.

This note provides that calibration. The contributions are:

- an exact minimum-weight decoding (MLD) oracle — exhaustive enumeration for $n \leq 13$ and integer programming (HiGHS MILP) for larger codes — against which every decoder failure is classified as avoidable or not;
- a small lemma: a syndrome with two minimal errors always carries a fractional point in the optimal face of the decoding LP, so face-fractionality detects *all* ambiguous syndromes;
- measured failure-mode tables for surface-code patches of distance 2–6 and the $[[72,12,6]]$, $[[144,12,12]]$ bivariate bicycle codes, at four channel rates with fixed seeds.

2 Setup and classifiers

We work on the X -error side of CSS codes: a check matrix h ($m \times n$, over $\mathbb{F}_2$) detects bit-flip errors e , produces the syndrome $s = he$, and a logical matrix ℓ ($k \times n$) maps an error to its logical class ℓe . The decoders are min-sum BP with OSD post-processing (BP+OSD, combination sweep) and BP alone with hard decisions; the MLD oracle enumerates, or solves by MILP, the minimum-weight errors for a syndrome.

- **Avoidable failure.** A decode of error e is an *avoidable failure* when the MLD oracle decodes the same syndrome successfully but the decoder fails logically ($\ell(e \oplus c) \neq 0$).

- **Degenerate ambiguity.** A syndrome is *ambiguous* when its minimal errors carry more than one logical class. A failure is attributed to degenerate ambiguity when its syndrome is ambiguous.
- **Stopping set.** A pattern is a stopping set when every check adjacent to its support touches at least two flipped positions. The test is applied to the true error support; on the residual it is vacuous for syndrome-valid corrections (every codeword support is trivially a stopping set).
- **Convergence.** BP-only output satisfying all checks is a *wrong convergence*; output violating the syndrome is a *non-convergence*.

3 An LP degeneracy detector

Gu and Soleimanifar [3] relax minimum-weight decoding of a syndrome s to a linear program over per-qubit indicators $x_i \geq 0$ coupled to per-check, parity-consistent subset weights; the objective $\sum_i x_i$ lower-bounds the minimal error weight. We call an optimum *fractional* when the *optimal face* — the set of optima, not one solver vertex — contains a point with a coordinate strictly between 0 and 1. The face is characterized exactly: fix the optimum value v^* , then bound each coordinate’s range on the face with $2n$ additional LP solves.

Lemma 1. *Let h be an $m \times n$ binary check matrix and s a syndrome with at least two distinct minimum-weight error patterns. Then the optimal face of the Gu–Soleimanifar LP for (h, s) contains a fractional point.*

Proof. Every error pattern e is feasible with objective $|e|$: for each check j the subset $S_j = N(j) \cap \text{supp}(e)$ has parity s_j , so setting the corresponding subset weight to 1 satisfies the parity and coupling constraints, giving $x = e$. Hence $v^* \leq d_s$, the minimal error weight. If $v^* < d_s$, no integral feasible point attains v^* (an integral feasible point is an error pattern of weight v^* , contradicting minimality), so every point of the face is fractional. If $v^* = d_s$, two distinct minimal errors $e_1 \neq e_2$ both lie on the face, and their midpoint $(e_1 + e_2)/2$ is feasible by convexity with objective d_s and fractional coordinates wherever they differ. $\square$

The lemma has two empirical consequences. First, every ambiguous syndrome is detected: recall is 1 *by construction*, not by luck. Second, the only false positives are multi-minimal-but- unambiguous syndromes — degeneracy without ambiguity — which is exactly the quantity this study measures. The single-solver-vertex signal is strictly weaker: HiGHS returned integral vertices for most degenerate syndromes, e.g. 0 of the 2 ambiguous syndromes of $[[5,1,2]]$.

4 Experiments

Two scales, all reproducible with fixed seeds:

- **Exhaustive** ($n \leq 13$): every one of the 2^n error patterns is decoded and classified against the brute-force oracle; every rate is exact.
- **Sampled** (Rivanna HPC): 10^4 shots per rate for the distance 4–6 surface patches (with the ILP oracle) and 10^5 shots per rate for the bicycle codes (decoder failure modes and LP calibration), at $p \in \{0.02, 0.05, 0.08, 0.10\}$.

The codes are the hypergraph product of two repetition codes (surface-code patches) and qudec’s bivariate bicycle constructions. Table 1 summarizes the test set.

5 Results

Table 2 reports the failure-mode split for the degenerate family at $p = 0.1$; Figure 1 shows the detector calibration across distances, and Figure 2 the avoidable share of BP+OSD failures.

Table 1: Codes under test. Instances: patterns (exhaustive) or shots per channel rate (sampled).

code	$[[n, k, d]]$	m	syndromes	instances	oracle	decoder
Steane	$[[7, 1, 3]]$	3	8	2^7	brute force	BP / BP+OSD
rep5	$[[5, 1, 5]]$	4	16	2^5	brute force	BP / BP+OSD
hgp22	$[[5, 1, 2]]$	2	4	2^5	brute force	BP / BP+OSD
hgp32	$[[8, 1, 2]]$	3	8	2^8	brute force	BP / BP+OSD
hgp33	$[[13, 1, 3]]$	6	64	2^{13}	brute force	BP / BP+OSD
hgp44	$[[25, 1, 4]]$	8	—	$10^4 \times 4$	ILP	BP / BP+OSD
hgp55	$[[41, 1, 5]]$	10	—	$10^4 \times 4$	ILP	BP / BP+OSD
hgp66	$[[61, 1, 6]]$	12	—	$10^4 \times 4$	ILP	BP / BP+OSD
bb72	$[[72, 12, 6]]$	36	—	$10^5 \times 4$	none	BP / BP+OSD
bb144	$[[144, 12, 12]]$	72	—	$10^5 \times 4$	none	BP / BP+OSD

Table 2: Surface patches at $p = 0.1$. Rates over all patterns (exhaustive rows) or 10^4 seeded shots; ambiguity share of avoidable BP+OSD failures; LP optimal-face precision/recall against avoidable failures.

code	$P(\text{fail})$ BP	$P(\text{avoid})$ BP	$P(\text{fail})$ OSD	$P(\text{avoid})$ OSD	ambig	prec	rec
hgp22	0.1800	0.1476	0.1800	0.1476	1.000	0.500	1.000
hgp32	0.2440	0.1978	0.2440	0.1978	1.000	0.494	1.000
hgp33	0.1576	0.0869	0.1520	0.0803	1.000	0.403	1.000
hgp44	0.183	0.130	0.149	0.088	1.000	0.214	1.000
hgp55	0.211	0.155	0.139	0.069	0.996	0.091	1.000
hgp66	0.246	0.196	0.143	0.072	0.996	0.091	1.000

5.1 Finding 1: degeneracy, not stopping sets

98–100% of avoidable BP+OSD failures are degenerate ambiguities on every degenerate code at every rate (Table 2); stopping-set failures contribute zero everywhere. On the non-degenerate controls the contrast is sharp: Steane has *no* avoidable failures at all (BP+OSD is MLD-optimal there), yet 3/8 of its syndromes show fractional LP faces — the signal fires and flags nothing.

5.2 Finding 2: the LP detector tracks degeneracy, not failure

Recall is 100% everywhere by Lemma 1; precision decays with distance (Figure 1) because large surface patches accumulate multi-minimal-but-unambiguous syndromes. The vertex signal would have missed most of this: 0 fractional vertices on $[[5, 1, 2]]$ and $[[8, 1, 2]]$, and 7 of 33 multi-minimal syndromes on $[[13, 1, 3]]$.

5.3 Finding 3: roughly half of BP+OSD failures are avoidable

At $p = 0.1$ the avoidable share of BP+OSD failures is 82% on the small patches and 50–59% on the distance 4–6 patches (Figure 2); the remainder are true MLD limits. The detector should therefore be read as a diagnosis tool for the recoverable half, not as a failure predictor.

5.4 Finding 4: the two bicycle codes fail differently

Table 3 and Figure 3 show the decoder-side calibration: on $[[144, 12, 12]]$ only 3% of BP failures are wrong convergences (BP simply does not converge on the failures it makes), while $[[72, 12, 6]]$ wrong-converges on 37% — a clean code-structure contrast. The LP precision against raw BP+OSD failures rises with p on both codes, from 0% to 78% on $[[144, 12, 12]]$, with recall 97–100% wherever labeled syndromes exist.

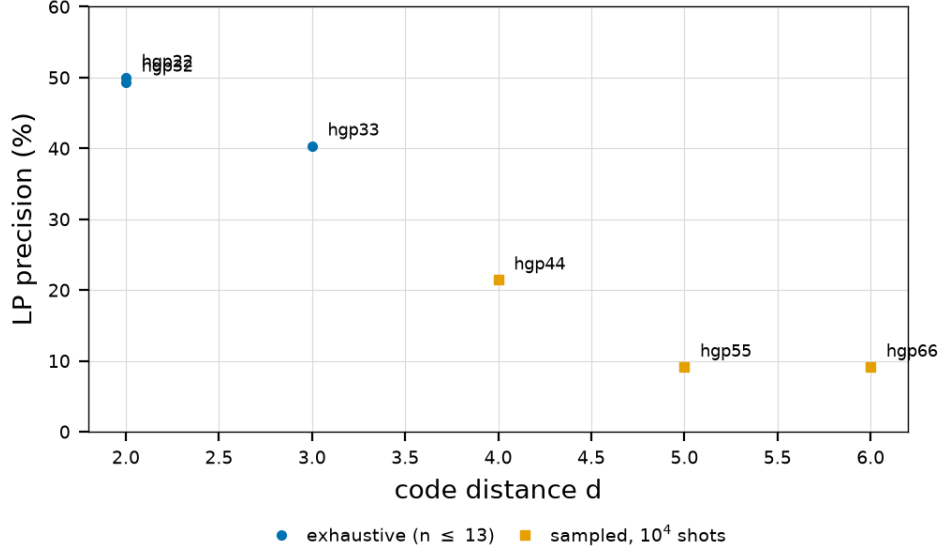


Figure 1: Fractional-LP precision against avoidable BP+OSD failures versus code distance, $p = 0.1$. Recall is 100% at every point (by Lemma 1). Precision decays as larger patches accumulate degenerate-but-unambiguous syndromes.

Table 3: Bicycle codes, 10^5 seeded shots per rate, no MLD oracle: LP optimal-face precision/recall against BP+OSD failures, and the wrong-convergence share of BP failures.

code	p	$P(\text{fail})$ BP	$P(\text{fail})$ OSD	wrong-conv	prec	rec
bb72	0.02	0.013	0.011	0.48	0.519	1.000
bb72	0.05	0.180	0.160	0.47	0.679	0.935
bb72	0.08	0.504	0.469	0.41	0.741	0.907
bb72	0.10	0.699	0.663	0.37	0.858	0.897
bb144	0.02	0.003	0.001	0.000	0.000	n/a
bb144	0.05	0.103	0.052	0.037	0.188	1.000
bb144	0.08	0.492	0.354	0.036	0.585	0.969
bb144	0.10	0.758	0.635	0.033	0.776	1.000

6 Limitations and outlook

- Code-capacity, X -error side only; one BP+OSD configuration (min-sum, 30 iterations, damping 0.9, OSD-CS). The rates are single fixed-seed batches; multi-seed error bars are the obvious next step.
- The LP face characterization costs $2n + 1$ LP solves per syndrome, which caps the bicycle-code calibration at a few hundred sampled syndromes; the ILP oracle is exact but bounded by per-solve time limits (zero timeouts occurred here).
- Ambiguity labels depend on the logical basis returned by the construction; the degenerate-family claims are robust to that choice.

References

- [1] S. Bravyi, A. W. Cross, J. M. Gambetta, D. Maslov, P. Rall, and T. J. Yoder, High-threshold and low-overhead fault-tolerant quantum memory, *Nature* **627**, 778–782 (2024).

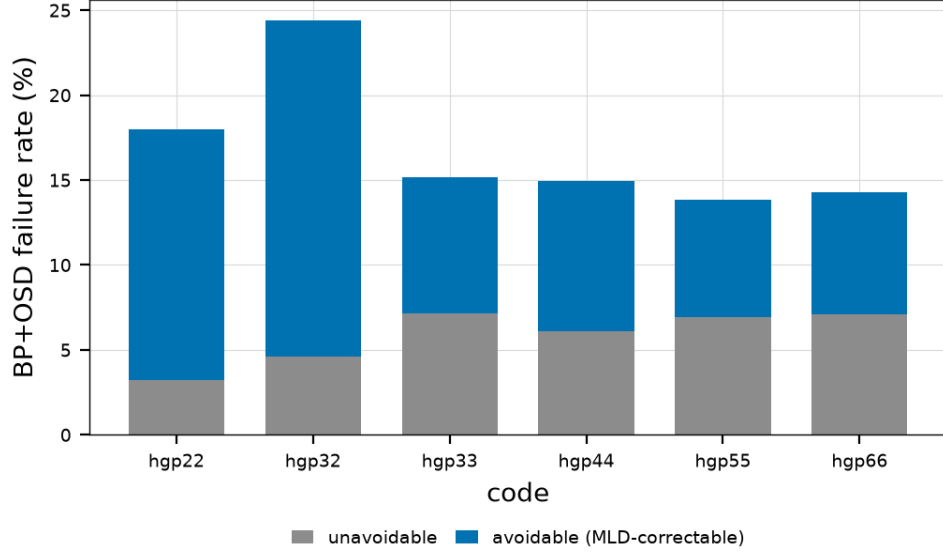


Figure 2: BP+OSD failure rate split into avoidable (the MLD oracle decodes the same syndrome) and unavoidable, $p = 0.1$.

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- [4] D. Poulin and Y. Chung, On the iterative decoding of sparse quantum codes, *Quantum Inf. Comput.* **8**(10), 987–1000 (2008).

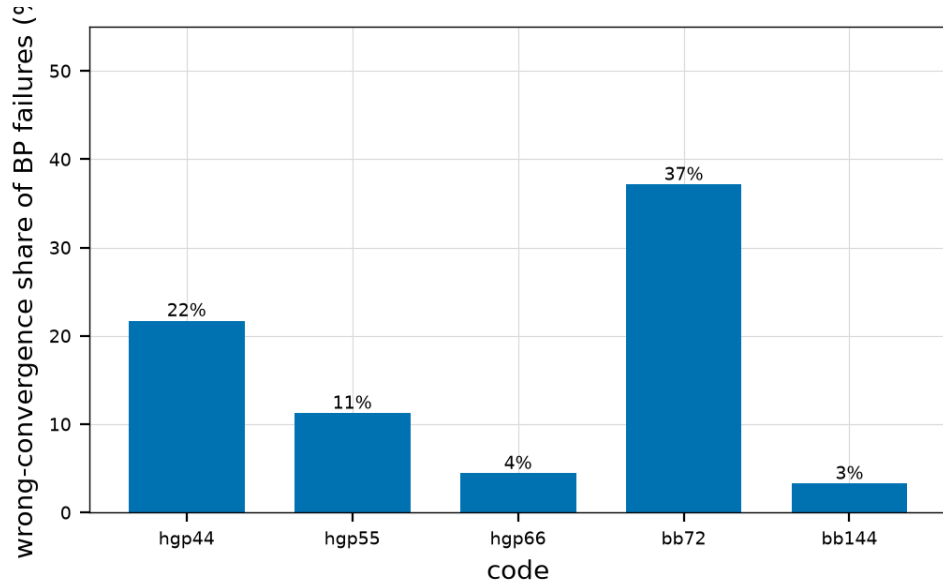


Figure 3: Wrong-convergence share of BP failures at $p = 0.1$ across the sampled codes (surface patches: avoidable failures; bicycle codes: all failures).

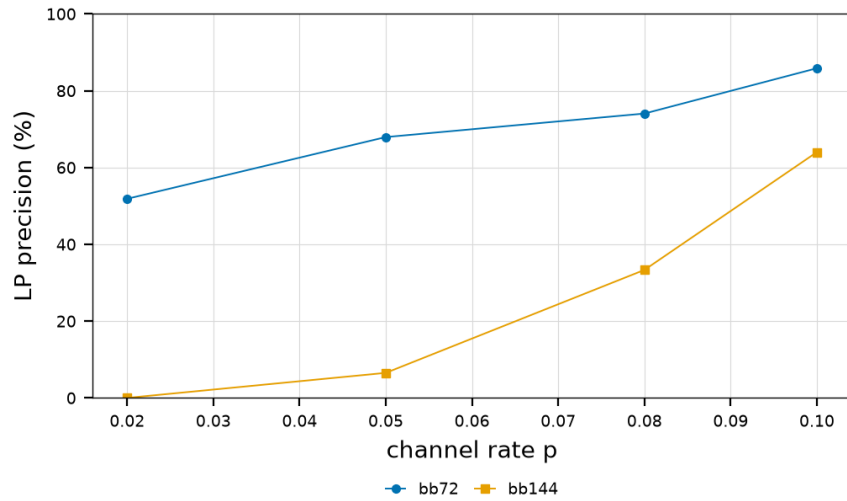


Figure 4: Fractional-LP precision against BP+OSD failures versus p on the bicycle codes.